

User Requirements Specification For The Lost & Found Item Tracker Serving Group 8

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1. Introduction

1.1 Objectives

This is the User Requirements Specification for the Lost & Found Item Tracker system. These requirements will ensure that the system will correctly and reliably perform its intended functionality.

The objective of the Lost & Found Item Tracker system is to create a web-based platform for reporting, tracking, and claiming lost items. The ability to connect users to their lost items must be a simple and fast process through an organized and searchable system.

1.2 Scope

This User Requirements Specification applies to the Lost & Found Item Tracker system. The URS will address the functions the system must be able to perform successfully to perform the primary functionality. This document also addresses the regulatory requirements for a system like the Lost & Found Item Tracker.

1.3 Assumptions

The application will be running on a properly functioning computer or mobile device, with a consistent network connection

1.4 Exclusions

This validation applies to the Lost & Found Item Tracker system and not to the various web browsers that may be used to access the application.

2. Program Requirements

2.1 Core Functional Requirements

- **PR-01:** The system shall allow a user to submit a found item through a form that collects details, including the item's title, description, location, date found, category, and an optional photo.
- **PR-02:** The system shall display all submitted items in a centralized, searchable list accessible to any authenticated user.
- **PR-03:** The system shall allow users to filter and search for items based on the following:
 - Category (default: *"all"*)
 - Location (default: *unspecified*)
 - Date found (default: *newest-first*)
 - Whether items have been marked as "claimed" (default: *disabled*)

- **PR-04:** The system shall allow an item's status to be updated to "claimed" once it has been returned to the owner, which will, in turn, remove the post from "default" public search results.

2.2 Usability Requirements

- **PR-05:** The item submission and search workflows shall be designed to require no more than three user actions from the main screen.
- **PR-06:** The system shall be responsive and optimized for mobile browsers, ensuring accessibility across devices.

2.3 Security Requirements

- **PR-07:** The system shall verify user authenticity through a valid login credential.
- **PR-08:** Sensitive information, such as user contact details and message contents, shall only be viewable by involved parties.
- **PR-09:** The system shall provide a means to expedite conflict reporting and/or resolution, wherein a "conflict" could include:
 - Incorrect or intentionally false claims of ownership
 - Deceitful "found" claims
 - User harassment

3. Data Requirements

3.1 Item Data

- **DR-01:** Each item record shall have a unique, system-generated Item ID that identifies it within the database.
- **DR-02:** Each record shall store the Item Title, Description, Category, Location, and Date Found or Date Lost.
- **DR-03:** Each record shall include a Status field (e.g., "Found," "Lost," "Claimed," "Archived") that reflects its current state.
- **DR-04:** The system shall store an optional photo for each item.
- **DR-05:** Each item shall store a Reporter ID referencing the user who created the listing to ensure accountability.
- **DR-06:** The system shall record timestamps for creation, last modification, and claim completion.
- **DR-07:** Claimed or inactive items shall remain accessible in an archived state for administrative review and audit purposes.

3.2 User Data

- **DR-08:** Each user shall have a unique User ID and authenticated University Email Address.
- **DR-09:** The system shall store the user's Display Name.
- **DR-10:** Each user shall have an assigned Role (e.g., Standard User, Administrator, Moderator) that governs access permissions.

- **DR-11:** The system shall track User Activity Logs for submission, modification, and claim events to support traceability.
- **DR-12:** Personally identifiable information, including names and emails, shall only be accessible to administrators and not displayed publicly without consent.

3.3 Communication Data

- **DR-13:** Each message shall have a unique Message ID and be associated with a Conversation ID between two users.
- **DR-14:** Each message shall include Sender ID, Receiver ID, Timestamp, and Message Content.
- **DR-15:** These records shall only be accessible to the involved users and administrators for moderation purposes
- **DR-16:** Notification records shall be stored when a user receives a message, claim update, or new item alert.

3.4 Data Integrity

- **DR-17:** All user-submitted input shall undergo server-side validation to ensure required fields are complete and contain valid data types.
- **DR-18:** The database shall enforce referential integrity between users, items, and messages
- **DR-19:** User authentication tokens or password hashes shall be stored using industry-standard encryption
- **DR-20:** The system shall maintain a consistent timestamp standard (e.g., UTC) across all data entries to ensure chronological accuracy.

4. Life Cycle Requirements

4.1 Updates and Scalability

- **LCR-01:** The system's architecture and database schema shall be designed to accommodate additional data fields and features without major redesign

4.2 Decommissioning

- **LCR-02:** If the system should be replaced or discontinued, a structured decommissioning process shall ensure safe removal of user data, following data protection standards.

5. Regulatory Requirements

In the event that the Lost & Found Item Tracker system is implemented beyond the student project and put into use by the student body at UW-Milwaukee, the following regulatory requirements should be implemented.

5.1 Privacy & Data Protection

In order for the system to comply with the FTC Act §5 against “unfair or deceptive practices”, the system would have to post a Privacy Policy that explains what personal data is collected by the system, why this data is collected, who is able to access the data, and how users can delete their data or close accounts. The data must also be stored securely with encrypted login information and limited access to user information.

5.2 Terms of Service (ToS) / User Agreement

In order to better protect the users of the platform, there should be a ToS Agreement created to define user responsibilities, ban illegal content or misuse of the system, and outline how to suspend or delete accounts that violate the terms listed. The ToS will be able to be upheld via a content moderation policy for uploaded images and descriptions to be reviewed if they violate the ToS or are images not allowed on the system.

5.3 Accessibility Requirements

The system should try to follow the Web Content Accessibility Guidelines version 2.1 (WCAG 2.1) for accessibility requirements. These include that alt text is added to the images of lost items, and the system uses sufficient color contrast (minimum 4.5:1 for text). The system should have form fields and buttons labeled clearly, and the text should resize well with changes to the window size and on mobile platforms.

6. Workflow

The Lost & Found Item Tracker will follow a straightforward workflow emphasizing ease of use and traceability. The main user interactions are described below.

6.1 Reporting a Found Item

1. The user navigates to the “Submit Found Item” page.
2. A form to collect item details, including a title, description, location, date found, category, and an optional photo, is displayed to the user.
3. Upon submission, the form's data is validated and stored in the database along with a generated item ID.
4. After a submission is completed, a confirmation message is displayed to the user, and the item is added to the searchable list.

6.2 Searching for Lost Items

1. The user navigates to the “Find Lost Items ” page.

2. A search interface listing “found items” is displayed to the user with filters for category, location, and date.
3. Users can select an item from the list to view more details, including the item description, image, and the means to contact and claim the item.
4. If an item's ownership is verified, the item's status is updated and marked as “claimed,” archiving the item and removing it from the public search list.

6.3 Claiming Items

1. When a potential match between a user and a found (lost) item in the system is found, the user can reach out to the finder using the internal messaging system.
2. The user can then send a claim request via the item’s page.
3. Once a user is verified as the owner of a found (lost) item, the item’s status is changed to “claimed,” archiving the item and removing it from the public search list.

6.4 Admin Oversight

- Admins and moderators can manage user permissions, review flagged content, examine archived data, and see system logs for submissions and claims for traceability and user disputes.

7. Stretch Goals

The following features are potential features that may be added to the Lost & Found Item Tracker in the future to further improve the user experience.

7.1 Lost-Item Alerts and Notifications for Potential Matches

Implementation of a background system that analyzes and matches users with potential lost (found) items based on criteria provided by the user. When similarities are detected between the user's query criteria and the description of a newly submitted item, the user will be notified within the app of a potential match.

7.2 Refined UI/UX for Desktop Experience

Implementation of a UI/UX designed for a desktop browser.

7.3 Campus Collaboration for Item Drop-Off and Retrieval

Partner with UWM’s lost and finds along with campus security to integrate the designated campus lost and found locations as drop-off and pickup zones, allowing for scheduled pickups and further verification of ownership of claimed items.

7.4 Image Matching to Suggest Similar Items

Implement AI image recognition that searches the images in the list of lost (found) items for items that are similar to a user input image to reduce manual search efforts.

8. Conclusion

The Lost & Found Item Tracker program proposed in this User Requirement Specification document is designed to provide a streamlined and user-friendly means of reporting, locating, and reclaiming lost items from within the UWM community in a stress-free and intuitive fashion. This document outlines and defines the Lost & Found Item Tracker program's necessary functional, data, and regulatory requirements to carry out the system's scalability, security, and user accessibility.

Our design specifications promote efficiency and accountability through secure user authentication as well as the implementation of administrative oversight to ensure that user disputes are properly handled. The planned stretch goals for possible future implementation, including notifications, improved desktop UI/UX, campus collaboration, and AI image matching, were conceived with the goal of further improving the efficiency with which users can find their lost items in order to create a better user experience.

Overall, this document acts as the foundation for the design, implementation, sustainability, and successful deployment of the Lost & Found Item Tracker program.

9. Citations

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Functionality Specification For The Lost & Found Item Tracker

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1. Functional Requirements

1.1 Data Requirements

The system must support structured and secure handling of item and user data in order to facilitate core lost and found functionality. The data that must be collected and processed includes item name, description, category, date found/lost, location, item status, and optional image for data relating to items, and user email address for data relating to users. It is important to note that no additional personal data should be stored in accordance with FERPA and data minimization principles recommended under GDPR.

All data captured by the system must be validated prior to storage. Item descriptions must not contain personal identifiable information. Date fields must not accept future dates. Uploaded files must adhere to approved format and size limits. These validation requirements align with the OWASP ASVS recommendations for secure data handling.

1.2 Operations Performed

The system shall support a defined set of operations that enable users to report, search for, and retrieve information about lost and found items in a structured and efficient manner. These operations shall be designed to ensure compliance with applicable privacy, security, and institutional regulations.

The system shall allow users to submit structured reports for either lost or found items. Each report shall include only non-personal information such as item type, description, location, time frame, and the optional media attachments. User-submitted data shall exclude personal information to maintain compliance with FERPA, GDPR, and university standards.

Users shall be able to browse a continuously updated list of reported items. The system shall support filtering by category, date, location, and item status. These operations shall be designed with efficiency and accuracy in mind.

The system shall provide item detail pages containing descriptions supplied by the reporting user. Contact mechanisms shall be designed using a secure message protocol or anonymized communication when applicable.

The system shall allow users to withdraw their submitted item listing. Data removal operations shall comply with data retention requirements applicable under university policy and GDPR, including the right to remove data from the database.

Users shall be permitted to update a previously submitted report. All updates shall be logged in accordance with university record-keeping requirements.

The system shall include the ability for authorized users to remove inappropriate content, flag policy violations, and address misuse as needed. Administrative actions shall be restricted to designated roles and shall follow OWASP access control principles.

1.3 Data Workflows

The system must implement data workflows governing the lifecycle of item information:

1. The item is submitted through a designated form.
2. The data is validated and stored in the database.
3. The item becomes accessible in the searchable item listing.
4. A user may request to claim the item.
5. The system archives the item after resolution or a set expiration time frame.

Because this system is used on the UWM campus, data retention must follow UWM IT governance and GDPR storage limitation principles, which require that item records be archived or deleted when no longer operationally necessary.

1.4 System Outputs

The system must produce the following outputs:

- Public Item Listings: filtered and searchable displays of active items, without displaying user data
- Administrative Reports: summaries of total items submitted, claimed, archived, and active, without displaying student information to maintain FERPA compliance.
- Audit Logs: internal logs documenting submissions, edits, and administrative actions, only made accessible to authorized personnel.
- Communication Logs: provides a record of all interactions between owners and reporters. Include timestamps for each message, the identities of the participants, and content type. Ensures transparency, supports dispute resolution, and maintains an auditable history of all communications related to the item's recovery process.
- Match Notification: Alerts and informs users of lost items that closely correspond to their own. Provides a brief summary of both items. Notification is designed to streamline the recovery process, quickly bringing possible matches to the owner.

1.5 Access Control Requirements

To properly delineate who enters data, the overall users include those that are responsible for reporting lost items, the claimants of lost items, administrators of the software, and potential moderators.

The standard users, provided that they have been authenticated, are allowed to submit a found item, where data including title, description, location, category, images, and private messages are viewed and utilized. Simply, the standard users are granted surface level view of the application, and only the user interface is available for alterations.

Moderators are granted higher permissions in areas that concern the maintenance of the application. Their roles include but aren't limited to answering user reports, reviewing false claims, investigating disputes, accessing flagged records, and moderation of user conflicts. Additional interface to maintain and the social and practical integrity of the application is assigned.

Administrators such as system owners, developers, and employees have access to the codebase and full permissions. Personal data of every user may be viewed and modified, sensitive data concerning the application are given, handling of legal obligations, and all of the above.

2. Interface Requirements

2.1 User Interface Requirements

The system should possess a main page that supports the core set of operations and views. These views must include a clean, responsive, and simple interface with options to allow specific actions such as taking the user to corresponding extended pages. These include searches of lost items based on filters, notifications, submission of items, and communication.

Users should be able to view and respond to chat messages through usual texting styles. The interface should also provide a dashboard showing a user's reported lost and found items and current interactions. The UI must comply with accessibility standards, including readable fonts, hints for images, and a possible screen reader compatibility. Most common image formats are to be supported for uploading and scanning, such as PNG, JPG, and JPEG.

2.1.1 Field Requirements

- Item Name: Accepts text input
- Description: Accepts text input
- Category: Select from a predefined list of categories
- Date Found/Lost: Only accepts dates on or before today
- Location: Accepts text input
- Status: The item's current state (Lost, Found, Claimed, Archived) defaults to Lost
- Item Image: Accepts jpg/png/jpeg file
- Search Field: Accepts text input
- Search Filters: Category, Date (searches dates on or after selected date and before today), Location, Status (won't appear in results if item is marked as claimed)

2.1.2 Screen Requirements

- Home: Displays links to Item Submission, Item Search, and Dashboard screens
- Item Submission: Displays required fields for submitting items (Item Name, Description, Ect.) and confirmation upon item submission
- Item Search: Displays required fields for searching, filtering searches, and a list of items based on search and filter criteria (Clicking item links to its Item Detail screen)
- Dashboard: Displays a list of items associated with the user's claims (Clicking item links to its Item Detail screen), a list of possible item matches based on users' "Lost" item submissions (Clicking item links to its Item Detail screen), and a list

of the user's Lost and Found Submissions (Clicking item allows user to remove or edit item information)

- Item Details: Displays required fields from Item Submission (Item Name, Description, Ect.) and a means to message the finder
- Admin: Only accessible by admin accounts and displays all, flagged, or archived items in order to moderate and remove content

2.1.3 Interaction/Accessibility Requirements

- Required fields must be filled out for completion of Item Submission
- User input fields provide inline error messages for invalid inputs
- Search results must update based on user input and filters

3. Business Requirements

3.1 Operational Rules and Data Dependencies

The system must enforce the following business logic to ensure consistent and compliant operations:

- Item data must be entered and validated before an item becomes visible in public listings.
- An authenticated user must initiate the claim workflow before an item can be marked as resolved.
- Administrators must be able to archive items that have been claimed or that have exceeded a designated retention period.
- The system must adhere to FERPA by ensuring no student-identifying information is displayed publicly or shared outside authorized workflows.
- The system must adhere to GDPR-informed principles by ensuring the minimization, purpose limitation, and timely deletion of stored user data.

4. Security Requirements

4.1 Client-Server Data Handling

The system shall utilize strategies to prevent common attack vectors:

- UI and database actions that have permission requirements will always have those requirements checked by the server
- Using strictly defined “prepared statements” in database interaction scripting to prevent SQL injection attacks
- All cross-interface variables will be sanitized prior to server-side handling to prevent XSS attacks

4.2 General Practice

The system shall apply a range of strategies recommended by OWASP:

- Only showing detailed error descriptions to users with Moderator-level permissions by using a generalized upper-layer of error handling
- Using HTTPS to encrypt all traffic between client and server
- Only asking for and storing user information crucial for the service's operation
 - An example of unnecessary information include a user's birthday
- Regular manual updates of dependencies aided by automatic checking
- Using test-cases meant to explicitly test security features that enforce, among others, the principle of least privilege
- Keeping a centralized log of activity to both prevent and learn from attacks

Citations

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Prior Art
For The
Lost & Found Item Tracker

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1. Snapchat Community Story

1.1 Snapchat Lost and Found Functionality

Snapchat is a multimedia messaging application that allows users to send photos, videos, and messages that disappear after being viewed or after a short duration, typically 24 hours after the initial posting. One common use of the platform within local or university communities has been for lost and found purposes, facilitated through the Snapchat Community Story feature. This functionality allows users that share a campus to post temporary stories viewable by other users, enabling others to see lost or found items in their vicinity.

The process is informal and user-driven: individuals who discover an item can upload an image and an optional short description to their story, while others who recognize the item are able to contact them through Snapchat's direct messaging system. The platform's emphasis on visual content and its large user base make it an accessible method for spreading awareness quickly within local areas.

However, this approach is unstructured, temporary, and dependent on user visibility. This solution is further hindered when taking into account the various other uses for the community Snapchat story, which can result in "flooding" of the feed, making it harder for users to find lost items. Snapchat stories expire after 24 hours, and there is no centralized database or search functionality for locating specific posts once they are removed. As a result, while Snapchat serves as a convenient communication tool, it lacks the infrastructure necessary for a dedicated, reliable lost and found tracking system.

1.2 Comparison to the Proposed System

Our Lost and Found Tracker system builds upon the concept of community engagement seen in Snapchat but addresses its core limitations. The proposed system will provide a centralized and persistent database where lost and found items can be stored indefinitely until claimed or removed by an administrator. Users will be able to add, browse, and search posts using filters such as item type, date, and location.

The platform aims to maintain the open, user-focused spirit of Snapchat by allowing individuals to post and find items directly, without the need for heavy moderation. However, unlike Snapchat, the information will remain accessible until an item is claimed, ensuring that users can return later to check or update listings.

1.3 Summary of Key Differences

Feature	Snapchat Community Story	Proposed Lost and Found Tracker
Data Persistence	Temporary (24-hour expiration)	Persistent, until item is claimed or removed
Search and Filtering	None	Search and filter by category, date, or location
User Verification	None	User registration for contact consistency
Moderation	None	Minimal, focused on user autonomy
Accessibility	Requires Snapchat app and enrollment in university community story	Accessible via web or mobile browser
Intended Use	General social interaction	Dedicated lost and found tracking

2. UW-Milwaukee Lost & Found Services

2.1 University Services Functionality

For those on the university grounds, the UW-Milwaukee University Police offer a variety of services, one of which being a lost and found service. According to the University Police website, their lost and found service is for retrieving lost items of value that were turned in around campus and brought to the officers. People who are attempting to find their lost property are directed to either visit the police department or call their non-emergency phone number to retrieve their missing property.

The university also has mention of lost and found services being offered through the service desk in the Student Union, as well as the service desks at each of the residence halls for those living on campus. These locations each have their own phone number to communicate with to access the lost and found services; however, there is no indication on any of the sites that there is any cooperation or synergy between locations.

2.2 Comparison to Proposed System

Our Lost and Found Tracker system is intended to be designed with centralization intended. The services offered through the university do not seem to coordinate throughout the six different phone numbers and service desks being utilized. By being web-based, our system will centralize the lost item postings to make finding an item easier for the user. The proposed system would also allow the users trying to locate their lost items to be able to browse and search without having to physically travel to the locations to try and find their items. The proposed system also allows users who are

more averse to making phone calls to search for and coordinate the retrieval of their missing items nonverbally through the system.

2.3 Summary of Key Differences

Feature	University Services	Proposed Lost and Found Tracker
Data Persistence	Temporary (24-hour expiration)	Persistent, until item is claimed or removed
Search and Filtering	Physically going to service desk or calling related phone numbers	Search and filter by category, date, or location
User Verification	None	User registration for contact consistency
Moderation	None	Minimal, focused on user autonomy
Accessibility	Requires ability to physically travel to service desks or access to a phone	Accessible via web or mobile browser
Intended Use	General social interaction	Dedicated lost and found tracking

3. ReclaimDesk

3.1 ReclaimDesk Functionality

ReclaimDesk is a web-based lost and found system designed for use by large organizations like airports, hotels, and other public venues. The website provides a centralized system for tracking lost and found items, including many of the same features that we intend to implement in our Lost and Found Item Tracker system. It works as an extension of a traditional lost and found, allowing the organisation's staff to log found items with descriptions and pictures, allowing users who have misplaced items to search for their item on ReclaimDesks database. If a user finds their lost item on their website, they can then send a claim request online via email. At this point, if the claim request is accepted, the user is able to schedule a time to come and pick up their item from the organization using ReclaimDesks system.

3.2 Comparison to Proposed System

Our Lost and Found Tracker is similar to ReclaimDesk, but functions at a smaller community level rather than at an industry level. Some of the features that are the same between the two are the submission of found items, which include descriptions, photos, location, and date. Both also provide the ability to search for and claim lost items. The main difference between ReclaimDesk and our proposed Lost and Found Item Tracker is

the fact that ReclaimDesk relies on locations where lost items are taken and then uploaded to the website via the staff of the business using their subscription service, while our Lost and Found Item Tracker is designed so that found items can be uploaded by either users who come across lost items or staff at one of the many decentralized lost and found locations on the UWM campus effectively centralizing the campus lost and found. The key difference with our approach is the fact that when scheduling a pickup, it would not necessarily be directly through one of the campus lost and found locations, and may instead be carried out via both the user who found the item and the user who lost it.

3.3 Summary of Key Differences

Feature	University Services	Proposed Lost and Found Tracker
Data Persistence	Persistent, keeps past items in the database for audits	Persistent, until item is claimed or removed
Search and Filtering	Advanced search with keyword and category matching	Search and filter by category, date, or location
User Verification	User registration via email	User registration for contact consistency
Moderation	Staff managed	Minimal, focused on user autonomy
Accessibility	Subscription-based web application (free for users)	Accessible via web or mobile browser
Intended Use	Dedicated lost and found tracking	Dedicated lost and found tracking

4. Survey Methodology

The analysis of existing solutions was conducted through a focused review of social media systems, app-based systems, and local systems that facilitate community-driven lost and found exchanges. The primary goal of this survey was to identify commonly used tools and assess their functionality, accessibility, and effectiveness in supporting item recovery within a university setting.

4.1 Snapchat Community Story

Snapchat was identified as a relevant platform for comparison due to its widespread use among university students for informal lost and found communication. Users frequently post images or messages about lost items to community stories, making the information visible to nearby users.

This assessment was based on targeted search engine research as well as personal experience with how students at the University of Wisconsin-Milwaukee utilize the

platform. The findings revealed an opportunity for improvement through a dedicated lost and found system with persistent, searchable records.

Search Engines Accessed: Google, Bing

Keywords Used: “Snapchat lost and found,” “Snapchat story lost item,” “Snapchat community story UWM,” “Snapchat found item post,” “Snapchat campus lost and found”

4.2 University Lost & Found Services

Services offered by UW-Milwaukee were identified as a relevant platform for comparison due to it being a service offered to students and visitors to the university. Information for the assessment was obtained through search engine research as well as communication with UW-Milwaukee’s University Police.

University Police Contact: Sergeant Joseph Kelsey

Search Engines Accessed: Google

Keywords Used: “UW-Milwaukee lost and found”

4.3 ReclaimDesk

ReclaimDesk was identified as a relevant platform for comparison due to it being a large-scale corporate-level implementation of a centralized web-based lost and found item tracker. Information for the assessment was obtained via ReclaimDesks website.

Search Engines Accessed: Google

Keywords Used: “ReclaimDesk”

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Technology Survey For The Lost & Found Item Tracker

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1. Frontend Selection & Justification

1.1 React JavaScript Library

React, also known as React.js or ReactJS, is an open-source front-end software library, with its primary programming language being JavaScript. React allows developers to build applications using components, which are small, reusable pieces of the user interface, of which can be buttons, forms, navigation bars, search bars, drop-downs, and any other constituents. Each component manages its own state and is able to combine to form complex interfaces, providing real-time updates to the application as data is modified during development.

1.2 Vue.js

Vue is a frontend framework that utilizes JavaScript for its functionality. Vue is marketed as an approachable, performant, and versatile framework that is scalable for web user interfaces. It was created by a former Google employee who wanted to improve upon the Angular framework Google developed taking the positives from Angular and building something lightweight.

Vue builds on top of standard HTML, CSS, and JavaScript with an intuitive API and documentation. It utilizes a reactive rendering system to be able to visually see how changes to the code impact the elements. Vue is also designed to scale with additional libraries and frameworks to better customize the system for the developer's needs.

1.3 Angular

Angular is an open-source JavaScript-based frontend platform that is supported by Google. It follows a Model-View-Controller architecture for designing a web application that allows each part to be developed in parallel and separately tested, allowing the code to be easier to understand and maintain.

Angular largely uses TypeScript, a syntactical superset of Javascript developed by Microsoft for large applications, instead of JavaScript. Angular is also designed with mobile browsers in mind to provide support for mobile web and native mobile applications.

1.4 Comparison Table

Software	Pros	Cons
React JavaScript Library	React's component-driven design encourages modular, reusable UI elements. This, in turn, further enhances code organization, consistency, reusability, scalability, and maintainability across larger projects. The technology has highly efficient performance with the Virtual Document Object Model as React's DOM minimizes costly direct interactions with the real DOM, of which only the necessary parts of the UI are re-rendered, improving application speed and responsiveness. React also has a strong ecosystem and community support, proven by its extensive documentation and with thousands of third-party libraries and developer tools. With an active community, it ensures quick issue resolution, frequent updates, and abundant learning resources.	While React itself is simple, setting up a full-featured app (with routing, state management, build tools, etc.) can involve complex configuration, which at times leads to developers needing to combine multiple libraries to achieve framework-level functionality. The React ecosystem evolves quickly, with certain third-party tools becoming obsolete or incompatible over time, requiring developers to stay updated with frequent library updates, new patterns, and caution for potential outdated documentation. JSX can be confusing or unappealing for developers who prefer separating HTML and JavaScript logic, adding a small learning overhead compared to template-based frameworks like Vue or Angular.
Vue	Vue is flexible and allows developers to use it as a library for specific features or as a full-fledged framework for complex applications. Vue allows for defining components in a single file, which includes the template, script, and styles, making it easier to manage components. Vue is known for its simplicity and approachability, making it easy for beginners to understand and use.	Vue is a relatively young technology compared to React and Angular, as such there are less experienced developers for the technology. Vue has a risk of over flexibility which could give a development team too many options that might lead to different programming approaches occurring within one team.
Angular	Angular is a full-feature framework with a comprehensive set of tools and libraries for	Angular is verbose and complex due to the way that components are managed, possibly requiring up to

Angular	building complex web applications. Angular follows the Model-View-Controller (MVC) architecture, which helps in organizing and managing code. Angular has a built-in dependency injection system that helps in managing dependencies and promoting modular code.	five files for a single component with the necessary dependencies. Angular has a difficult learning curve compared to React or Vue, as one would have to learn TypeScript in addition to the modules, dependency injection, and components of Angular.
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1.5 Final Selection

The team has decided to utilize React for the frontend framework, as members involved in its development are most familiar due to prior experience with the software. With React's reusable structure, encapsulation, vast ecosystem of high-quality third-party libraries, and extensive documentation, this approach enhances code and architecture comprehension, consistency, and long-term maintainability throughout the development of the Lost and Found Item Tracker application.

2. Database Selection & Justification

2.1 PostgreSQL

PostgreSQL is a powerful, open-source relational database management system (RDBMS) known for its strong adherence to SQL standards, full ACID compliance, and extensive feature set. It supports advanced data types and robust indexing options, making it ideal for complex applications that require flexibility and reliability. PostgreSQL integrates exceptionally well with Django, offering built-in support for features not found in other database management systems.

From a development perspective, PostgreSQL offers powerful tools for migrations and backup management, both of which are useful in a collaborative project setting. Its active community and detailed documentation make it a dependable choice for long-term maintenance. The primary drawbacks are slightly higher setup complexity and resource consumption compared to lighter databases such as MySQL or SQLite. However, its reliability, standards compliance, and compatibility with Django make it the preferred option for scalable web applications such as the Lost & Found Item Tracker.

2.2 MySQL

MySQL is one of the most widely used open-source databases and is well-known for its speed, simplicity, and ease of deployment. It is a popular choice for web applications due to its straightforward configuration, efficient query performance for read-heavy workloads, and strong support across major hosting services. Django supports MySQL natively, and it performs well for typical applications with moderate complexity.

MySQL has several limitations, however, that can impact long-term scalability and data integrity. MySQL's query planner and indexing system are efficient for common queries but less flexible for complex joins or analytical workloads. Additionally, some advanced Django features, like partial indexes and native JSON operations, are limited in MySQL compared to PostgreSQL.

Despite these drawbacks, MySQL remains a reasonable choice for lightweight or mid-scale web applications where ease of setup and fast read performance are higher priorities than strict standard compliance or advanced data modeling.

2.3 Oracle

Oracle Database is an enterprise-grade RDBMS designed for large-scale applications that demand high reliability, extensive optimization capabilities, and advanced security features. It offers powerful transaction handling, fine-grained access control, and performance tuning tools that make it suitable for mission-critical systems. Oracle provides strong support for SQL standards, as well as integration options for large enterprise infrastructures.

However, Oracle's commercial licensing model and high administrative complexity make it impractical for most academic or small-team projects. Setting up and maintaining an Oracle instance requires more resources and expertise compared to open-source alternatives. Although Django includes official support for Oracle, it is rarely used in educational or prototype environments due to the high cost and heavy configuration overhead.

For a project such as the Lost & Found Item Tracker, Oracle would be excessive in both scale and cost, offering little practical advantage over open-source systems like PostgreSQL that already meet the project's needs for reliability, consistency, and scalability.

2.4 Comparison Table

Database	Pros	Cons
PostgreSQL	PostgreSQL offers high support in Django, as many Django features are only fully supported in PostgreSQL. PostgreSQL also features strong data integrity and reliability. PostgreSQL supports complex queries and powerful indexing PostgreSQL is highly scalable, offering excellent performance for large datasets and multiple-user workloads.	PostgreSQL is more complex than other considered options, which can make the barrier for entry higher for group members with less experience in database management. PostgreSQL has higher memory usage, which might not be suitable for small-scale database designs.
MySQL	MySQL has a long history of use in web applications, and is stable and reliable. MySQL has very good performance for simple queries, making it ideal for smaller scale projects. MySQL is easy to set up and is versatile, being supported by many different hosting providers.	MySQL has limited feature support for Django, which can limit the operations done on the database. MySQL has weaker data integrity than other considered options, and might not be as trustworthy for larger scale projects.
Oracle	Oracle is highly robust with excellent scalability, promising strong performance. Oracle has excellent reliability and security features, ensuring strong data integrity and security. Oracle comes with official Django backend support, allowing it to integrate well into web applications.	Oracle features a complex setup, requiring extensive configuration and licensing. Oracle can be costly, as it is a commercial software that comes with licensing fees. Oracle can be overkill for smaller projects such as academic or small-team projects.

2.5 Final Selection

After evaluating PostgreSQL, MySQL, and Oracle, the team selected PostgreSQL as the database technology for the Lost & Found Item Tracker project. PostgreSQL provides the best balance of reliability, scalability, and compatibility among the options considered. Its advanced feature set, such as native support for JSON data and robust indexing capabilities, aligns well with the project's need for accurate item tracking, search

functionality, and multi-user interaction. Additionally, PostgreSQL's open-source nature and active community ensure both accessibility and long-term maintainability. While MySQL offers simplicity and performance for smaller-scale applications, it lacks some of the advanced capabilities needed for future feature expansion. Oracle, though powerful, is cost-prohibitive and unnecessarily complex for this scope. Overall, PostgreSQL offers the most practical and technically sound foundation for building a reliable, scalable, and maintainable web application.

3. Backend Selection & Justification

3.1 Django

Django is an open-source, high-level Python Web Framework widely used for building secure and scalable web applications. It follows the Model-View-Template (MTV) architecture and includes built-in tools for tasks like authentication, database management, and form handling, allowing developers to create robust applications quickly. Django's strong emphasis on security helps protect against common web vulnerabilities such as SQL injection and cross-site scripting.

Django's modular structure and object-relational mapping (ORM) system enables efficient data management and seamless integration with databases like PostgreSQL. Using our Lost & Found Item Tracker project, for example, Django would simplify the implementation of features we will add to ensure data consistency and reliability. Its well organized codebase and extensive documentation makes the application simple to maintain, test, and debug.

3.2 Node.js

Node.js is an open-source JavaScript runtime environment well known for its use in the development of servers and web applications. It is easily scalable due to its non-blocking, event-driven architecture, which makes it useful for handling simultaneous connections with real-time data processing. Node.js also has existing React integration along with PostgreSQL integration libraries that would allow for the quick and easy implementation of secure data management and features like user authentication.

From a development perspective, the non-blocking, event-driven architecture of Node.js would ensure efficient performance while carrying out data-heavy operations in our Lost & Found Item Tracker, such as item claim updates and item searching. Node.js would also be very convenient for the application's maintainability and debugging, since it has the benefit of using JavaScript for both the frontend and backend development.

3.3 Comparison Table

Framework	Pros	Cons
Django Web Framework	Its speed and scalability allows it to build and scale up projects incredibly quickly which is why it is popular in tech startups. Django has the best sets of documentation among open source frameworks out there. With well organized, beginner friendly, and practical examples. Django has a built-in security that prevents it from common threats and attacks such as XXF, CSRF, and SQL injections, as well as HTTPS. Django has great versatility so it is compatible with a wide range of applications from content management systems or an entire social network. It can even carry out machine learning or data analysis.	Django's high-level framework is made for more complex projects and applications. Simpler ones are often better off built with simpler frameworks such as Flask. Large scale projects with many database interactions may lead to slower performance if not optimized properly. Before building the code developers must build the whole structure out due to its complexity. Django does not have any conventions, a rulebook of sorts, which can lead to slower developments or less standardized.
Node.js	Node.js's non-blocking event-driven architecture is beneficial for scalability and the handling of simultaneous connections with real-time data processing. Node.js allows for full-stack JavaScript development, handling both the front and backend with JavaScript. Node.js's default package manager provides over 2 million module libraries, which would allow for fast and easy integration of features like authentication.	Node.js is not optimal for heavy CPU tasks since it operates on a single-thread event loop, which could impact performance if improperly handled. The asynchronous programming model that Node.js provides could result in deeply nested callbacks if unfamiliar with the framework, impacting the code's readability and maintainability. Node.js's rapid ecosystem can cause its module libraries to become outdated or

		deprecated which could result in surprise problems with previously functioning code.
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3.4 Final Selection

After evaluating Django and Node.js, the team has decided to work with the Django Web Framework for the Lost & Found Item Tracker. Much of the team has prior experience with Django and is familiar with its built-in tools, which will streamline the implementation of core requirements like user verification and form handling. Django is also compatible with PostgreSQL and can be easily integrated to work with React via the Django REST framework, allowing for the quick and reliable development, scalability, and maintainability of the Lost & Found Item Tracker application.

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